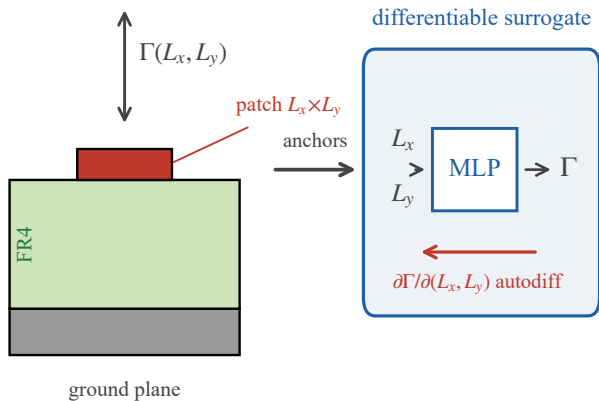


(a) unit cell + differentiable surrogate

waveguide simulator ($PMC \perp x$, $PEC \perp y$)



(b) full-wave reflection-phase surface

